

Nasif Fahmid Prangon

PhD Researcher, Computer and Information Sciences
Temple University, USA

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Research Interests

Edge Intelligence, Distributed Learning Systems, Knowledge Distillation, Federated Learning, Network-Aware Machine Learning, Cloud and Edge Computing

Education

Temple University, USA

Ph.D. in Computer and Information Sciences

Advisor: Dr. Jie Wu

Aug 2023 – Present

CGPA: 3.83/4.00

Rajshahi University of Engineering and Technology (RUET), Bangladesh

B.Sc. in Electronics and Telecommunication Engineering

2014 – 2018

CGPA: 3.44/4.00

Publications

Beyond the Teacher–Student Paradigm: The Role of Network Structure in Knowledge Distillation

N. F. Prangon and J. Wu

International Conference on Computing, Networking and Communications (ICNC), 2026

Key Idea: Topology-aware knowledge distillation showing how structures like hypercube, ring, and chain impact learning efficiency in distributed systems.

Resource Optimized Task Offloading in Delay Sensitive Edge Networks

N. F. Prangon, A. Sawwan, and J. Wu

International Conference on Computing, Networking and Communications (ICNC), 2025

Key Idea: Energy and delay-aware task allocation for edge networks.

AI and Computing Horizons: Cloud and Edge in the Modern Era

N. F. Prangon and J. Wu

Journal of Sensor and Actuator Networks (JSAN), 2024

Key Idea: Survey of distributed AI and cloud-edge intelligence.

Research Experience

Graduate Research Assistant

Aug 2023 – Present

Temple University

- Conducting research on the impact of topology and network structure for distributed edge AI systems.
- Designing communication-efficient learning frameworks using structured network topologies.
- Working on LLM applications in healthcare (Summer 2024, Summer 2025).

Teaching Experience

Teaching Assistant

Aug 2023 – Present

Temple University

- Computer Systems and Low-Level Programming (C, memory, pointers, OS concepts)

Industry Experience

Key Account Manager (Wholesale)

Axiata Group, Malaysia

Oct 2022 – Jul 2023

- Conducted large-scale traffic analysis and capacity planning across international telecom networks, collaborating with multiple OpCos to forecast demand and optimize service delivery.
- Managed roaming and interconnect services (IPX, SCCP, SIGOS), ensuring reliability and performance while evaluating cost-performance trade-offs for efficient resource allocation.

Specialist, Roaming Network (Core Network Planning)

Robi Axiata Limited

Aug 2022 – Jul 2023

- Designed and evolved scalable roaming network architectures, translating business requirements into deployable technical solutions through vendor evaluation and system integration.
- Led capacity planning and network expansion initiatives, coordinating cross-functional teams to maintain performance, reliability, and service quality under increasing traffic demands.

System Engineer (Core Network Operations)

Novotel Ltd

Feb 2020 – Jul 2022

- Managed telecom core infrastructure including SIP signaling, SBCs, and SMS gateways, ensuring high availability through monitoring, troubleshooting, and performance optimization.
- Implemented routing optimizations and maintained carrier integrations (SMPP, HTTP, SS7), leveraging tools such as Wireshark, tcpdump, MySQL, and Elasticsearch for traffic analysis and issue diagnosis.

Technical Skills

Programming: Python, C, SQL, MATLAB

Systems & Tools: Linux, Wireshark, tcpdump, MySQL, Elasticsearch, Kibana

Networking: SIP, SMPP, IP Routing, Core Network Systems

ML/AI: PyTorch, Scikit-learn, Transformers, Time Series Models

Projects

Twitch Friendship Prediction

[Repo Link](#)

- Built a link prediction pipeline on social graph data to predict streamer-user relationships using node features, graph metrics (e.g., Adamic-Adar), and embedding methods.
- Evaluated multiple models (Logistic Regression, XGBoost, SVM) and analyzed the impact of graph-aware features on prediction performance.

MindGuide: Intelligent Mental Health Platform

[Repo Link](#)

- Developed an AI-assisted platform for mental health guidance, integrating structured data extraction and analysis of online therapy resources.
- Designed pipelines to evaluate service quality, privacy compliance, and tracking behavior across multiple mental health websites.

EduQuery: Intelligent Academic Query System

[Repo Link](#)

- Built a query-driven system to assist users in retrieving academic information using structured data and intelligent search mechanisms.
- Designed backend workflows for processing user queries and organizing educational resources efficiently.

Languages

English (Professional), Bangla (Native)